

Math 526, F08
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	15	
3	15	
4	15	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3})$.

(a) Find the length $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{3^2 + (\sqrt{3})^2 + (2\sqrt{6})^2} = \sqrt{9 + 3 + 24} = \sqrt{36} = 6$$

$$\|\mathbf{w}\| = \sqrt{0^2 + (-\frac{\sqrt{3}}{3})^2 + (\frac{\sqrt{6}}{3})^2} = \sqrt{\frac{1}{3} + \frac{2}{3}} = \sqrt{1} = 1$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{3 \cdot 0 + \sqrt{3} \cdot (-\frac{\sqrt{3}}{3}) + 2\sqrt{6} \cdot \frac{\sqrt{6}}{3}}{6 \cdot 1}$$

$$= \frac{-1 + 4}{6} = \frac{3}{6}$$

$$= \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, \sqrt{3}, 2\sqrt{6})}{6} = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$$

(d) Choose a number c so that $2\mathbf{w} + c\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$(2\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$$

$$(2\mathbf{w}) \cdot \mathbf{v} + (c\mathbf{v}) \cdot \mathbf{v} = 0 \Rightarrow 2(\mathbf{w} \cdot \mathbf{v}) = -c\|\mathbf{v}\|^2$$

$$c = \frac{-2(\mathbf{w} \cdot \mathbf{v})}{\|\mathbf{v}\|^2}$$

$$= \frac{(-2) \cdot 3}{6^2}$$

$$= -\frac{1}{6}$$

Problem 2. (15 points)

- (a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ x - y \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

- (b) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 2 and 3, then rows 1 and 2.

$$\mathbf{P}_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \mathbf{P}_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{P} = \mathbf{P}_{12} \mathbf{P}_{23} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- (c) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A}^2 = \mathbf{0}$ but $\mathbf{A} \neq \mathbf{0}$.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Problem 3. (15 points)

(a) Write down the inverse of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ if it is invertible.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(b) Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find A^{-1} using the Gaussian-Jordan Elimination.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\ell_2 = 1} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \left. \vphantom{\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}} \right\} \text{Gaussian}$$
$$\xrightarrow{\ell_3 = 2} \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\ell_{13} = 2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{array} \right] \left. \vphantom{\begin{array}{ccc|ccc} 1 & 1 & 0 & -3 & 0 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{array}} \right\} \text{Jordan}$$
$$\xrightarrow{\ell_{12} = 1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \end{array} \right] = [I : A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Problem 4. (15 points) Let $A = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix}$.

(a) Find the LU factorization of A .

$$A = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix} \xrightarrow{\ell_{21} = \frac{1}{3}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ -2 & 4 & -1 \end{bmatrix}$$

$$\xrightarrow{\ell_{31} = -\frac{2}{3}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 6 & \frac{20}{3} \end{bmatrix}$$

$$\xrightarrow{\ell_{32} = 3} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix} = U$$

$$\Rightarrow A = LU$$

where $L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{3} & -3 & 1 \end{bmatrix}$ $U = \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix}$

(b) Find the LDU factorization of A using the result from (a).

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{3} & -3 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & \frac{20}{3} \end{bmatrix} \quad U = \begin{bmatrix} 1 & 1 & \frac{4}{3} \\ 0 & 1 & -\frac{5}{6} \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 5. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 2 & 5 \\ 1 & 2 & 3 & 7 \\ 2 & 4 & 6 & 14 \end{bmatrix}$.

(a) Transform A to its ordinary echelon form U using Gaussian elimination.

$$A = \begin{bmatrix} 1 & 2 & 2 & 5 \\ 1 & 2 & 3 & 7 \\ 2 & 4 & 6 & 14 \end{bmatrix} \xrightarrow{e_2 = e_1} \begin{bmatrix} 1 & 2 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 2 & 4 & 6 & 14 \end{bmatrix}$$

$$\xrightarrow{e_3 = e_3 - 2e_1} \begin{bmatrix} 1 & 2 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\xrightarrow{e_3 = e_3 - 2e_2} \begin{bmatrix} 1 & 2 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

(b) What is the $\text{rank}(A)$? What are the free variables?

$$\text{rank}(A) = 2$$

Free variables: x_2, x_4

Pivot variables: x_1, x_3

(c) Find a special solution of $Ax = 0$ by for each free variable and then write down the complete solution.

i) set $x_2 = 1, x_4 = 0$, solve $Ux = 0$

$$\begin{cases} x_1 + 2x_2 + 2x_3 + 5x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 + 2x_3 = -2 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2 \\ x_3 = 0 \end{cases}$$

$$\Rightarrow S_1 = (-2, 1, 0, 0)$$

ii) set $x_2 = 0, x_4 = 1$, solve $Ux = 0$

$$\begin{cases} x_1 + 2x_2 + 2x_3 + 5x_4 = 0 \\ x_3 + 2x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 + 2x_3 = -5 \\ x_3 = -2 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \\ x_3 = -2 \end{cases}$$

$$\Rightarrow S_2 = (-1, 0, -2, 1)$$

$$\text{The complete solution: } x = c_1 S_1 + c_2 S_2 = c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

where c_1, c_2 any constants.

Problem 6. (15 points) True or False (no explanation needed).

(a) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 4 & 12 \\ -1 & -2 & -4 \end{bmatrix}$, there isn't any 3×3 matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.

True

(b) The set of all vectors (b_1, b_2, b_3) with $b_1 \leq b_2 \leq b_3$ is a subspace of $\mathbb{R}^3$.

False

(c) If a subspace of the vector space $\mathbf{M}$ contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix $\mathbf{I}$.

True

(d) A square matrix has at least one free variable if it is not invertible.

True

(e) Given an $m \times n$ matrix $\mathbf{A}$ and its reduced row echelon form $\mathbf{R}$, we have the nullspaces $\mathbf{N}(\mathbf{A}) = \mathbf{N}(\mathbf{R})$.

True